A manual for installing FASTCORE, rFASTCORMICS_v2, COBRA Toolbox V3, GUROBI & MATLAB R2024b for Windows 11

Windows version: 10.0.19042 Build 19042

1- Installing MATLAB R2024b

1. Go to https://nl.mathworks.com/ and create an account using your student/academic email.
2. Sign-in to the same website using your account.
3. Open https://nl.mathworks.com/downloads/ then click on "Show more" on the left side.
4. Choose R2024b version then download MATLAB installer.
5. Check these MATLAB toolboxes during installing:

Bioinformatics Toolbox (required)
Curve Fitting Toolbox (required)
Optimization Toolbox
Parallel Computing Toolbox
Signal Processing Toolbox
SimBiology
Statistics and Machine Learning Toolbox
Symbolic Math Toolbox

2- Installing GUROBI Optimizer :

2.1. Downloading GUROBI
2.2. Go to the GUROBI Download Center: https://www.gurobi.com/downloads/
Register and log in
2.3. Get a free license here: https://www.gurobi.com/academia/academic-program-and-licenses/
Save this license key, you will need it later.
2.4. Download the GUROBI Optimizer (version 12.0.3)
Execute the Gurobi-<ver>-win64.msi file and follow the installation instructions.
Accept the default path C:\gurobi<ver>\.
2.5. In order to set the environment variables on Windows, please follow the instructions here: https://support.gurobi.com/hc/en-us/articles/360060996432-How-do-I-install-Gurobi-on-Windows-without-administrator-credentials.
2.6. Browse to Start > All Programs > Gurobi < VER> > Gurobi Interactive Shell (<ver>). This should prompt to enter the license key you saved in 2.1.3. Enter this key and hiter ENTER.
2.7. Close the Gurobi Interactive Shell and open it again. This command should give you the prompt for Gurobi. Exit by entering exit() or hitting CTRL-D on your keyboard.

3- Install the latest git for Windows: https://git-scm.com/download/win

4- Installing COBRA Toolbox V3:

4.1. Run MATLAB as administrator

4.2. Double check that the setfield command has only this path "/Applications/MATLAB_R2024b.app/toolbox/matlab/datatypes/setfield.m" by running:

```
> which -all setfield
```

If there are other paths, remove them such as these examples:

```
> rmpath('C:\Program Files\MATLAB\R2019b\toolbox\finance\ftseries\')
> rmpath('C:\Program Files\MATLAB\R2019b\toolbox\sl3d\sl3d\')
```

4.3. In your command window, type the following lines (one by one):

```
> cd C:\gurobi1200\win64\matlab
> gurobi_setup
> savepath
```

4.4. Download COBRA Toolbox from https://github.com/opencobra/cobratoolbox by opening Git Bash, and then running

> git clone --depth=1 https://github.com/opencobra/cobratoolbox.git cobratoolbox

4.5. In MATLAB, add COBRA Toolbox folder to MATLAB paths by:

Clicking on Set Path
Clicking on Add with subfolders
Browsing to find your COBRA Toolbox folder, and then clicking on Select Folder
Clicking on Save

4.6. In MATLAB, navigate through your "cobratoolbox" folder.
Then run the following line in your command window:

```
> initCobraToolbox.m
```

You should see this line:

```
gurobi        active        1        1      1      1      -      -      -
```

4.7. To select GUROBI as your solver, run this line in your command window:

```
> changeCobraSolver('gurobi')
```

Note: If MATLAB crashes and closes during installation, open MATLAB and run again the `initCobraToolbox.m` and you might want to double-check that cobratoolbox is still added to the path using which `-all initcobratoolbox`. If the command is not found, repeat step 3.4.

5- Installing FASTCORE and rFASTCORMICS_v2:

5.1. From Git Bash:

> git clone https://github.com/sysbiolux/rFASTCORMICS

Or manually:

from « Code », click on "Download ZIP"

5.2. From MATLAB command window:

```
addpath(genpath('/PATH OF THE DOWNLOADED rFASTCORMICS/'))
```

5.3.A version of FASTCORE already exists in the COBRA Toolbox. However, we will use an updated version from the rFASTCORMICS github. We need to remove the one of the COBRA Toolbox from the path:

Use your Windows file explorer to find the FASTCORE folder corresponding to the following path:
'PATH\cobratoolbox\src\dataIntegration\transcriptomics\FASTCORE'
Delete the folder

***Warning**: In case you prefer not to delete the FASTCORE folder directly from MATLAB, you can also remove it from the MATLAB path. However, note that this step will need to be repeated each time you initialize the COBRA Toolbox.*

5.4. If you are able to call the following functions (and more related functions) in the command window of MATLAB then the installation is done.

- fastcc *(from the rFastcormics_v2/scripts/fastcore folder)*
- fastcore *(from the rFastcormics_v2/scripts/fastcore folder)*
- rFastcormics4cobra_v2 *(from the rFastcormics_v2/scripts/rfastcormics folder)*

Problem: COBRA toolbox is not detecting git during installation

1.1 Go to Edit the System Environment Variables (Quickly: enter 'env' in the windows search bar)
1.2 Click on Environment Variables...
1.3 In the first box named System variables, open “PATH”, then add these 2 paths:
    c:\Program Files\Git\bin\git.exe
    c:\Program Files\Git\cmd
1.4 Click ok to save the variable.

Problem: COBRAToolbox folder is detected to be owned by Git
Error message must contain:

- fatal: **detected dubious ownership** in repository at 'C:/Users/cobratoolbox/cobratoolbox-master'

Solution: Run the following code in Git bash

- git config --global --add safe.directory C:/Users/cobratoolbox/cobratoolbox